

World-Building Design Curriculum — Instructional Analysis & Build Plan

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Executive read

What you have is a strong, distinctive content inventory for a postgraduate-level program in world-building design practice; it is not yet a curriculum, and the fastest route to one is to treat Track 02's Concept DNA as the spine and rebuild everything else around it.

What the material is. Six tracks, 39 modules and roughly 250 named units. The through-line is a governing artifact, the Concept DNA (Track 02), which must survive translation into four practice studios (Track 03: art, architecture, object, fashion), be executed with computational tools (Track 04), and survive money, law and public release (Tracks 05–06). Track 01 supplies the foundations (history, form, senses, material, systems). The whole thing is deliberately anchored in the UAE, Gulf, Arab world, MENA and SWANA, which no comparable online program currently does.

What it reads like. A hybrid of a studio-based MA (RCA, Central Saint Martins, Design Academy Eindhoven, Bartlett, Parsons Transdisciplinary Design), a strategy-consultancy operating system (brand architecture, DNA, contributor briefs) and a founder curriculum (capital, IP, cash flow). That hybrid is its selling point and its main production risk: three audiences, three registers.

Diagnosis of the syllabus as written

Issue	Where	Why it matters	Fix
Topics, not outcomes	Every unit	"Mastering...", "the rigorous..." describe ambition, not what a learner can do or show	Rewrite each unit as 1–3 observable outcomes + one evidence type (see §2, §9)
No learner, level, time or modality	Whole document	You cannot size a module, price it or choose interactives without these	Decide the five questions below before module design

Issue	Where	Why it matters	Fix
Near-verbatim duplication	5.8 $\approx$ 6.1; 5.10 $\approx$ 6.4 + 6.5	Two capstones; learners meet the same text twice	Merge Tracks 05 and 06 into one Venture & Release track (§7)
Lifecycle/circularity taught six times	1.5, 3.3.10, 3.4.12, 4.9, 5.9, 6.3	Repetition without progression	One horizontal Planetary Weight thread with a level per track
Ergonomics/affordance taught three times	1.3, 3.3.2, 3.4.2	Same	Foundation in 1.3, applied per studio
Tech-pack/handover taught three times	3.2.10, 3.4.11, 6.2	Same	One Handover Grammar unit with studio-specific packs
Track 03 has no header and only four studios	Track 03	1.1 promises graphic, moving image, performing arts, digital culture; no studio delivers them	Add at least Brand/Graphic and Digital Product/Interface studios, or scope them out explicitly (§11)
Track 04 is eleven tools, each a career	Track 04	Learners will drown; a LIDAR unit and a drawing unit are not equal	Split into Core Toolkit (5) and Elective Deep Tools (6) (§6)
Register	Everywhere	Dense adjectival prose ("brutal", "uncompromising", "lethal") reads as manifesto, not instruction	Keep the voice for marketing and briefs; write learning copy plain
"Modular" for "Module"; a stray "qnot" in 3.1	Headings	Small, but signals a draft	Copy-edit pass

What is already excellent and should be protected. The Concept DNA method, with its constants/variables/thresholds, the four boundary tests (belongs, edge, failure, seductive-but-off-DNA) and the blind-interpretation test, is a teachable, assessable,

proprietary method. The mandatory adjacent-practice translation in 2.4 is the best single assessment idea in the document. The regional centering is a real market position, not decoration.

Decisions to make first

1. Learner and level: early-career designers, studio founders, brand and cultural-sector leads, or students? Postgraduate-equivalent is implied throughout; confirm it.
2. Product shape: one long program, or stackable products (10-hour micro-courses → 40-hour modules → 120-hour studios → diploma)? Stackable is recommended; it lets you ship Track 02 first.
3. Modality: self-paced, cohort, or hybrid? Studio pedagogy needs live critique; recommended is self-paced interactives plus fortnightly live crits plus an AI critic between them.
4. Physical making: kits and field assignments (Gulf sites, archives, workshops) or digital-only? The studios are hollow without making; budget for it.
5. Language: English-first with Arabic parallel, or bilingual by design? 5.7 already treats bilingual architecture as content; the program should practise what it teaches.
6. Pilot: build Module 2.2 (Concept DNA) first as a four-week micro-course with the DNA Builder interactive (§8, §13); it is the spine, it is self-contained, and it proves the method.

Program architecture

Restructured, the program is five levels and roughly 720 learner hours: Foundations (100 h) → Method (160 h, one continuous project) → one primary Studio plus one adjacent translation (160 h) → a Core Toolkit threaded through all of it (112 h + 32 h electives) → Venture & Release with a defended capstone (about 150 h). That is a postgraduate-diploma-sized program; the same modules also sell individually.

Module types used below. F = Foundation (knowledge and analysis), M = Method (procedural, project-based), S = Studio (making and critique), T = Tool (skills, drills), V = Venture (decision and simulation). Hours are learner hours (content + practice + critique), not production hours.

Track 01 — Context & Core Fundamentals (Level 1, 100 h)

Module	Type	Hrs	Prerequisite	Primary evidence	Change
1.1 Historical, Cultural & Critical Frameworks	F	24	none	Annotated counter-history map of one regional lineage + an ethics-of-reference case ruling	Split ethics/attribution into its own unit; it recurs in 3.x and 5.3
1.2 Morphology, Typology & Formal Logic	F	20	none	Typology dismantling study (one archetype, five subversions, one kept identity)	none
1.3 Somatic, Optical & Interaction Dynamics	F	20	1.2	Affordance audit + palette-and-light system with cultural coding notes	Make this the single home of ergonomics; studios apply it
1.4 Materiality, Semiotics & Textual Systems	F	20	1.2	Material-as-text dossier (five materials, lifecycle, semiotics, patina) + bilingual text architecture sketch	none
1.5 Systems Thinking & Planetary Scale	F	16	1.4	Ecosystem map + one scenario set for a chosen intervention	Make this level 1 of the Planetary Weight thread

Track 02 — Generative World Systems (Level 2, 160 h, one project)

Module	Type	Hrs	Prerequisite	Primary evidence	Change
2.1 Phase I Research, Narrative & World Premise	M	30	Track 01	Research synthesis board + World Premise v1 (stance, contradiction, non-goals, hypotheses)	none
2.2 Phase II Concept DNA & Generative Systems	M	40	2.1	Concept DNA v1.0 (constants, variables, thresholds, grammars, anti-DNA) + boundary-test set + blind-interpretation results	Pilot module; build first
2.3 Phase III World & Collection Architecture	M	30	2.2	Collection map (roles, families, drops, carriers) + risk/cost distribution	none

Module	Type	Hrs	Prerequisite	Primary evidence	Change
2.4 Phase IV Applied World Construction	M	60	2.3	Anchor expression + adjacent-practice translation + revision history + DNA v2.0	Runs as the mini-capstone; the full capstone in Level 5 scales it

Track 03 — Practice Studios (Level 3, 120 h primary + 40 h adjacent)

Studio	Type	Hrs	Prerequisite	Primary evidence	Change
3.1 Art, Installation & Experimental Practice (8 units)	S	120	2.2	Site-responsive installation proposal, simulated (light/sound/scale), with documentation-as-artwork plan and funding defence	none
3.2 Architecture, Space & Experience (10 units)	S	120	2.2	Programmed, choreographed spatial proposal: plan/section/sequence, atmospheric strategy, feasibility sheet, handover schedule	Fold 3.2.10 tech-pack into a shared Handover Grammar unit
3.3 Object, Product & Furniture (10 units)	S	120	2.2	Resolved object family: buck → prototype → BOM → compliance check → end-of-life plan	Fold 3.3.10 into Planetary Weight thread
3.4 Body, Fashion & Wearables (12 units)	S	120	2.2	Capsule (hero/core/carry-over), pattern + drape simulation, tech-pack, grading logic, campaign frame	Fold 3.4.11 and 3.4.12 into shared units
Adjacent translation (any second studio)	S	40	primary studio	One artifact of the same DNA in a second medium, critiqued against the DNA	New; formalises 2.4's requirement

Track 04 — Technical & Computational Studios (threaded, 112 h core + electives)

Module	Type	Hrs	Prerequisite	Primary evidence	Change
4.1 Analog Intelligence & Notational Logic	T core	20	none	Orthographic set + movement notation of one existing object or space	Runs alongside Track 01

Module	Type	Hrs	Prerequisite	Primary evidence	Change
4.3 Algorithmic Geometry, Scripting & Parametrics	T core	24	4.1	Parametric definition where one variable regenerates a family	Runs alongside 2.2
4.4 Synthetic Workflows & Machine Collaboration	T core	20	2.2	Prompt architecture + agent brief that reproduces the DNA under blind interpretation	Runs alongside 2.2; the two are one experience
4.5 Spatial Computing, Simulation & Interface	T core	24	4.3	Real-time walkthrough or interactive prototype of the studio project	Runs alongside Track 03
4.8 Optical Physics, Sonic Engineering & Visual Evidence	T core	24	4.3	Rendered and scored evidence set with a representation-ethics statement	Runs alongside Track 03
4.2 Reality Capture & Reverse Engineering	T elective	16	4.1	Point cloud or photogrammetry capture of a real site or artifact	Elective
4.6 Information Architecture & Data Visualization	T elective	16	2.1	Interactive visualisation of research data	Elective
4.7 Physical Computing & Mechatronics	T elective	16	4.1	Sensing/actuating prototype (kit-based)	Elective
4.9 Environmental Analytics & Lifecycle Computation	T elective	16	1.5	LCA and climatic simulation of the studio project	Elective; level 3 of Planetary Weight
4.10 Digital Fabrication & Material Mechanics	T elective	16	4.3	Fabricated component with toolpath and failure analysis	Elective
4.11 Data Architecture, Cyber-Ethics & Version Control	T elective	16	4.4	Versioned DNA repository with ethics review	Version control belongs in core (2.2); ethics stays elective

Tracks 05 + 06 merged — Venture, Release & Defence (Level 5, about 150 h)

Module	Type	Hrs	Prerequisite	Primary evidence	Change
5.1 Masterbrand Strategy & Orbital Architecture	V	12	2.3	Brand architecture map for the learner's world	none

Module	Type	Hrs	Prerequisite	Primary evidence	Change
5.2 Capital Structuring & Creative Finance	V	12	none	18-month cash-flow model with two funding scenarios	none
5.3 Legal Architecture, IP & Cultural Sovereignty	V	10	1.1 ethics unit	IP and attribution protocol for the project	Absorbs 6.2's IP bullet
5.4 Agency Operating System & Studio Governance	V	12	4.4	Studio OS (folders, agents, templates) + pitch structure	none
5.5 Omnichannel Commerce & Spatial Retail	V	10	3.x	O2O journey map + flagship-as-media brief	none
5.6 Community Governance & Cult Brand Dynamics	V	8	5.1	Access-tier and ritual design	none
5.7 Narrative-Led Campaigns & Activation	V	12	5.1	Bilingual campaign architecture + activation workflow	none
5.8/6.1 Implementation Logistics & Collaboration	V	12	3.x	Collaboration matrix + MVI trade-off record + phased drop plan	Merged
5.9 Venture Economics & Expansion	V	10	5.2	Margin architecture + one-market expansion plan	none
6.2 Technical Handovers & Compliance	V	12	3.x	Complete handover package audited by a peer	Shared Handover Grammar unit
6.3 Responsible Release & Lifecycle Operations	V	8	4.9 or 1.5	Release, maintenance and end-of-life plan	Level 4 of Planetary Weight
5.10/6.4/6.5 Systemic Transfer, Critical Defence & Dossier	V capstone	40	everything	Transfer brief (DNA applied to an unfamiliar constraint) + live defence + professional dossier	Merged into one capstone

Sequencing rule. Every learner carries one project from 2.1 to the capstone. Track 01 modules are taken before or alongside 2.1; Track 04 core modules are scheduled next to the module they serve (shown in the Change column); Track 05/06 modules are taken after the primary studio, so the money and law questions are asked about a real artifact, not an abstract one.

Track 01 — Context & Core Fundamentals

Track 01 is where online programs lose people, because it is where lecture-shaped content hides; the rule for this track is do first, then read, with every module opening on a lab or a field prompt in the learner's own city.

1.1 Historical, Cultural & Critical Frameworks

Dimension	Content
Essential question	Whose history is our design history, and what do we owe the sources we draw from?
Sub-questions	How are dominant timelines made and what do they exclude? What does a Gulf, Egyptian or Levantine modernity look like when not measured against Europe? Where exactly is the line between reference, quotation, adaptation, translation and appropriation? When is consultation and cultural authority required, and how is it documented? Who did the work (craftspeople, technicians, uncredited communities) and how do we attribute it?
Knowledge required	Movements and counter-movements; regional chronology (pre-oil, oil, post-oil; the Nahda; Cairo, Beirut and Baghdad modernisms; Gulf urban modernity from the 1950s; today's institutions); historiography and decolonial critique; source criticism, provenance research, oral-history interviewing

Dimension	Content
Resources needed	Archnet / Aga Khan Documentation Center; Arab Image Foundation; Barjeel Art Foundation; Sharjah Art Foundation and Art Jameel publications; scholars such as Nasser Rabbat, Yasser Elsheshtawy, Todd Reisz (Showpiece City), Ahmed Kanna; Escobar (Designs for the Pluriverse), Tunstall (Decolonizing Design), Said (Orientalism); image rights for a regional image bank; oral-history partnerships with craft communities; museum partners (Louvre Abu Dhabi, Etihad Museum, Zayed National Museum)
Teaching method	Comparative timeline building; case-based seminars; attribution-spectrum case rulings (learner rules, expert ruling revealed with rationale); one oral-history micro-fieldwork assignment; reading circles with a critique protocol
Interactives	Counter-Narrative Timeline (drag lineages, toggle dominant/obscured layers); Reference-to-Appropriation Spectrum (case cards placed on a scale, then compared with expert and community rulings); Obscured-Labour Credit Map (deconstruct a famous object into its contributors); Regional Material-Culture Atlas
Evidence	Annotated counter-history map of one regional lineage; three written case rulings; a draft attribution and consultation protocol carried into 5.3

1.2 Morphology, Typology & Formal Logic

Dimension	Content
Essential question	What makes a form hold together at any scale, and how far can you push an archetype before it stops being itself?

Dimension	Content
Sub-questions	What is the structural DNA of a chair, a garment, an interface, a room? How do grids and modules generate hierarchy? How do figure and ground work across 2D, 3D and digital? How do weight, rhythm and asymmetry produce equilibrium or tension? How do Islamic geometric systems operate as generative grids rather than decoration?
Knowledge required	Proportion, modularity, tessellation; gestalt; typology theory (Rossi, Moneo); scale and its cues
Resources needed	Ching (Form, Space and Order); Elam (Geometry of Design); Müller-Brockmann (Grid Systems); Broug and Critchlow on Islamic geometry; an archetype image bank (chairs, garments, rooms, interfaces) cleared for teaching use
Teaching method	Constrained iteration (100 variations in one hour); inductive form-finding; timed sketch drills; side-by-side critique
Interactives	Typology Dismantler (a parametric chair, garment or room with sliders for its structural DNA and an identity meter that shows where peers stopped recognising it); Grid Lab (generative grids over the learner's own composition); Visual Gravity game (place elements, see weight and tension scored); Scale-Agnostic Morph (one form rendered as object, room, city block)
Evidence	Dismantling study: one archetype, five subversions, one identity kept

1.3 Somatic, Optical & Interaction Dynamics

Dimension	Content
Essential question	How does a design tell the body what to do, and how do colour, light and time shape what people feel?
Sub-questions	What anthropometric and cognitive limits apply? How do affordances communicate without instruction? How does colour behave optically and how is it coded regionally (white kandura, black abaya, indigo, saffron, the greens)? How do palettes and lighting create hierarchy and depth? How do thresholds and sequences choreograph a journey?
Knowledge required	Albers's colour interaction; colour science basics (LAB, metamerism, colour temperature, CRI); Gibson and Norman on affordance; Fitts's law and cognitive load; Hall's proxemics; Cullen's serial vision; Tschumi's sequence
Resources needed	Albers; Norman (Design of Everyday Things); Hall (The Hidden Dimension); Cullen (The Concise Townscape); Pallasmaa; anthropometric datasets (Dreyfuss/Tilley); a regional colour-coding study, which does not exist yet and would need commissioning
Teaching method	Perception labs (do, then explain); micro-experiments on the learner's own objects; affordance audits in the field; storyboarding
Interactives	Colour Interaction Lab with cultural-coding overlays; Light Logic Sandbox (three.js scene, place lights, watch hierarchy and depth change); Affordance Guess-and-Test (predict use from an image, then see the reveal); Threshold Sequencer (storyboard compression and expansion on a timeline); Interface Cognitive-Load Meter

Dimension	Content
Evidence	Affordance audit of five artifacts; a palette-and-lighting system with cultural notes

1.4 Materiality, Semiotics & Textual Systems

Dimension	Content
Essential question	What does a material, a surface or a word say before it does anything?
Sub-questions	How do you read a material as cultural text (provenance, labour, ecology)? How do finish and patina steer the eye and age over time? How do signs trigger memory and expectation? How is text a structural material, and what does a bilingual Arabic-Latin architecture demand?
Knowledge required	Material properties and lifecycle basics; semiotics (Peirce, Barthes); typography, including Arabic script logic (kashida, calligraphic styles, bilingual pairing, mirrored layouts); patina and wear
Resources needed	A regional material archive (palm and areesh, sadu weaving, coral stone, gypsum naqsh, imported teak, contemporary composites) — likely a commissioned asset; Barthes (Mythologies); Forty (Objects of Desire); Chapman (Emotionally Durable Design); Lupton (Thinking with Type); Khatt Foundation and type designers such as Nadine Chahine for Arabic typography; bilingual signage case studies (Dubai Metro, Louvre Abu Dhabi, Expo 2020)
Teaching method	Material forensics (trace one object's supply chain); semiotic swap exercises; typography drills; before/after patina studies

Dimension	Content
Interactives	Material Card Deck with lifecycle and semiotic sliders; Patina Time Machine (a surface at 1, 5 and 20 years); Sign-Swap (change a material or symbol, watch the reading change); Bilingual Text Architecture tool (pair Arabic and Latin hierarchy, test mirroring)
Evidence	Material-as-text dossier (five materials); a bilingual text architecture for the learner's project

1.5 Systems Thinking & Planetary Scale

Dimension	Content
Essential question	What does this intervention touch that you did not design, and what happens to it in ten years?
Sub-questions	How do you map the ecosystem an intervention enters? What roles do algorithms and autonomous agents play as actors? What is the lifecycle and ecological weight? How do you generate futures scenarios that are useful rather than decorative?
Knowledge required	Stocks, flows and feedback loops (Meadows); actor-network thinking (Latour); LCA basics; foresight methods (scenario matrix, futures wheel, causal layered analysis); Gulf specifics — water, heat, import dependence, labour flows, the post-oil transition
Resources needed	Meadows (Thinking in Systems); Bratton (The Stack); Crawford (Atlas of AI); Dunne and Raby (Speculative Everything); Candy and Dunagan on experiential futures; Ellen MacArthur Foundation circular-design material; regional climate, water and energy datasets
Teaching method	Causal-loop mapping workshops; futures wheel; scenario writing; consequence audits of the learner's own previous work

Dimension	Content
Interactives	Ecosystem Map Builder (causal-loop diagram tool); Futures Wheel and Scenario Generator (variables in, four scenarios out); Planetary Weight Calculator, level 1 (rough LCA); Non-Human Actor Simulator (take the algorithm's point of view)
Evidence	Ecosystem map and one scenario set for a chosen intervention; this becomes the first entry in the learner's Planetary Weight thread

Track 02 — Generative World Systems (the spine)

Track 02 is the product: teach it as one continuous 16-week project in which the Concept DNA is a versioned document (v0.1 to v2.0) that every later track edits, and grade generativity (can others build from it?) rather than beauty.

The Concept DNA needs a schema before it can be taught or built as software.

Proposed structure, drawn only from what the syllabus already names:

1. Premise — proposition, stance, central contradiction, human scenario, non-goals
2. Constants (unbreakable), Variables (adaptable), Thresholds (maximum deviation before the concept breaks)
3. Grammars — formal, material, construction, chromatic, spatial, atmospheric, sensory, behavioural, interaction, temporal, narrative, verbal; each marked required, optional or not applicable for this world
4. Anti-DNA — what the system is incapable of being
5. Alignment — relationship to Practice DNA, Collection DNA and Masterbrand DNA
6. Category obligations — functional duties by product or experience type
7. Test set — one exemplar each of belongs, edge, failure and seductive-but-off-DNA
8. Contributor brief and agent prompt architecture — the blind-interpretation package
9. Version history — what changed, why, and which evidence forced it

Suggested cohort rhythm. Weeks 1–3 Phase I, weeks 4–7 Phase II, weeks 8–10 Phase III, weeks 11–16 Phase IV; two live crits per phase; an AI critic available between crits; one peer blind-interpretation exchange per phase.

2.1 Phase I — Research, Narrative & World Premise

Dimension	Content
Essential question	What do we know, what do we believe, and what will this world fight for?
Sub-questions	Which research routes (primary, secondary, archival, speculative) does the question actually need? How credible and how biased is each source? When has the work crossed the insight threshold, from clusters to a claim? What is the stance and the productive contradiction? For whom, in what scenario? Which archetypes and rituals define the world? What does it enable, protect, resist, reject, transform, and what are the non-goals?
Knowledge required	Research design and ethics (consent, attribution from 1.1); sensemaking (affinity clustering, insight statements); narrative structure (stakes, archetypes, ritual); world-building practice from film and games; design fiction
Resources needed	Kolko (Exposing the Magic of Design); Kumar (101 Design Methods); Wolf (Building Imaginary Worlds); the World Building Institute's approach (Alex McDowell); Superflux and Near Future Laboratory on design fiction; interview and archive-access training; a consent and attribution template
Teaching method	Two-week guided field research sprint; synthesis workshop; a rubric-gated insight threshold (no premise without a defended insight); premise writing with critique

Dimension	Content
Interactives	Research Synthesis Board (cluster, tag by source type and credibility, with an insight generator that forces the form "we believe X because Y, which means Z"); Source Credibility Grader; Stance and Contradiction Generator (tension pairs); World Perimeter Card (enable/protect/resist/reject/transform sliders plus non-goals); Hypothesis Log carried through all four phases
Evidence	World Premise v1 (two pages) and the synthesis board

2.2 Phase II — Concept DNA & Generative Systems

Dimension	Content
Essential question	Can someone who has never met you build the right thing from your rules alone?
Sub-questions	What is immutable, what flexes, and how far? Which of the twelve grammars does this world need, and which are irrelevant? What is the anti-DNA? How does this DNA sit against practice, collection and masterbrand DNA? What are the category obligations? Can you produce one artifact for each of the four boundary tests? Does the DNA survive blind interpretation by a peer and by an AI agent, and what do the misreads reveal?
Knowledge required	Rule design and emergence (from game design); design-system thinking (tokens and constraints as analogy); pattern languages; brand codes; semantic versioning; prompt architecture
Resources needed	Alexander (A Pattern Language); Salen and Zimmerman (Rules of Play); a small library of exemplary grammars (brand and design-system documents, with permissions); a regional exemplar set; model access for agent runs

Dimension	Content
Teaching method	Rule-writing labs; test-set production (four artifacts, one per category); blind-interpretation rounds (peer swap, then agent run); versioning discipline; critique on generativity
Interactives	Concept DNA Builder (structured editor on the schema above, with sliders for thresholds, grammar coverage radar, export as contributor brief and as agent prompt, version diff); Boundary Test Arena (sort outputs into the four categories, peers sort too, a disagreement heatmap shows where thresholds are fuzzy); Blind Interpretation Lab (DNA in, eight generated outputs out, learner rates adherence, misreads point to weak rules, edit, rerun; adherence charted across versions); DNA Diff Viewer
Evidence	Concept DNA v1.0, the test set, a blind-interpretation report and a change log

2.3 Phase III — World & Collection Architecture

Dimension	Content
Essential question	What is the system of things, and which thing goes first?
Sub-questions	What thesis governs this collection inside the larger arc? Which outputs are anchor, hero, core, entry, experimental, supporting? What rules govern families, series, editions and drops? How does coherence hold from environment to artifact without cloning? How are cost, complexity and risk distributed? What gets added, subtracted, edited, rejected? Which carriers, interfaces and channels does the world need?
Knowledge required	Range planning and merchandising logic; product-line architecture; editorial curation and pacing; risk distribution; brand architecture (ties to 5.1)

Dimension	Content
Resources needed	Range-plan templates; collection case studies with clear role logic (a fashion house's line system, a homeware range, a device family; regional luxury and modest-fashion brands); curatorial texts (Obrist, Ways of Curating)
Teaching method	Matrix workshops; a mock range review with a buyer; a forced-subtraction curation exercise; carrier selection against channel realities
Interactives	Collection Matrix Builder (roles by families with cost, risk and accessibility heat overlays); Drop Sequencer (timeline with an audience-momentum curve); Cross-Scale Coherence Checker (artifacts from micro to macro checked against the grammars); Cutting Room (subtract with justification)
Evidence	Collection map, risk and cost distribution, editorial calendar

2.4 Phase IV — Applied World Construction & Development

Dimension	Content
Essential question	Does the world survive contact with matter, bodies and other makers?
Sub-questions	Which primary practice and which anchor? Which adjacent translation, and what does it break? What construction logic? Which human scenario grounds it? Which prototype answers which question, at what fidelity? What was abandoned and why? Which specialists are needed? What changed in the DNA and what forced the change?
Knowledge required	Bridges into Tracks 03 and 04; prototyping strategy (each prototype as a question); critique; documentation of alternatives

Dimension	Content
Resources needed	Studio faculty and critics; prototyping budgets or kits; access to makerspaces and workshops (Tashkeel-type facilities); specialist consultation hours (engineer, fabricator, developer, artisan)
Teaching method	The integrated brief; weekly crit; a prototype-question log; an abandoned-directions archive; scheduled specialist consultations
Interactives	Prototype Question Planner (question, fidelity, medium, cost); Revision History Viewer (DNA versions against artifacts on one timeline); Abandoned Directions Archive; Adjacent Translation Challenge (a second medium drawn at random, then a DNA-adherence critique); Specialist Gap Finder
Evidence	Anchor expression, differentiated family, adjacent translation, evidence compilation, visible revision history, DNA v2.0

Track 03 — Practice Studios

All four studios share one shape — brief → research → iterate → crit → refine → document — and one online problem: making cannot happen on a screen, so each studio needs a real site, base-build or body, a physical kit, a simulation layer for what the kit cannot do, and fortnightly live critique with an AI critic between sessions.

Shared studio pattern. A 12-week studio: weeks 1–2 site and lineage, 3–6 concept and first prototypes, 7–10 resolution and simulation, 11–12 handover pack and defence. Async pin-up gallery with a structured feedback protocol (what it is trying to do, what it is doing, what to change); one live crit every two weeks; one specialist Q&A per studio (engineer, fabricator, pattern cutter, sound designer). Grading is against the learner’s own DNA first, then craft, then feasibility.

3.1 Art, Installation & Experimental Practice

Dimension	Content
Essential question	When utility is suspended, what is the work responsible to: the site, the audience, the institution, or the planet?
What is really being taught	Site reading, the ethics and mechanics of audience complicity, light/sound/matter as structure, durational thinking, and how to defend non-commercial work to funders
Online strategy	A site-responsive proposal for a real regional site (a warehouse in Alserkal Avenue, a courtyard in Sharjah's arts area, a desert site, a museum foyer), simulated in three.js and sound tools, with a 1:20 maquette kit; the documentation plan is graded as part of the work
Resources	O'Doherty (Inside the White Cube); Kwon (One Place After Another); Bishop (Installation Art; Artificial Hells); Bourriaud; Krauss; Schafer (The Soundscape); Oliveros (Deep Listening); Blesser and Salter (Spaces Speak); Careri (Walkscapes); regional case studies from the Sharjah Biennial, Diriyah Biennale, Desert X AIUla and Expo pavilions; guest critics from regional practice (artists such as Zeinab Alhashemi, Shaikha Al Mazrou, Lawrence Abu Hamdan, Monira Al Qadiri, Dana Awartani are the kind of voices to approach)
Evidence	Site-responsive proposal with light and sound simulation, audience-journey rules, decay or de-install plan, ecological ledger, and a mock funding defence

Unit	Teach as	Interactive
3.1.1 Lineage, theory, mythos	Case dissections; a critique of the desert-sublime trope	Container Neutraliser (drop one work into white cube, black box, street, desert; watch meaning shift); Lineage Map
3.1.2 Conceptualisation	Translation drills (poem to volume, census to material); deductive vs inductive sprints	Translation Engine (map data fields to spatial variables); What-If Constraint Generator
3.1.3 Scale and intervention	Site reading; scale studies against a human figure	Scale Power Slider (awe to intimacy in 3D); Site Reading Checklist
3.1.4 Light, chroma, matter	Light-as-mass labs; unstable-matter case studies	Light-as-Mass Sandbox (haze, projection, refraction); Ganzfeld Simulator; Volatility Timeline (ice, rust, growth)
3.1.5 Sound	Listening walks; soundscape capture; spatial audio basics	Sonic Boundary Mixer (place sources, walk an avatar, infrasound explained); Anechoic vs Reverberant A/B
3.1.6 Synthetic realities	Phone-AR overlay prototypes; live-data parameters; a post-human audience brief	Live-Data Installation Sim (weather or market feed drives form); AR Overlay Template
3.1.7 Relational, durational	Rules-of-engagement design; decay schedules; documentation planning	Audience Ritual Builder (approach, participation, exit); Ephemerality Planner; Archive-as-Artwork template
3.1.8 Impact and economics	Grant and patronage simulation; circular logistics; right-to-the-city case	Sponsor Negotiation Simulator (defend the DNA against a brand's asks); Ephemeral Debt Calculator

3.2 Architecture, Space & Experience

Dimension	Content
Essential question	How does a space make people move, feel and behave, and can it be built, permitted, paid for and maintained?
What is really being taught	Programming before form, spatial choreography, atmosphere as physics, adaptive reuse, and the invisible systems (BOH, egress, MEP, money) that decide whether a concept survives; codes are taught as literacy, not licensure — position the studio as spatial and experience design, not architectural qualification
Online strategy	One real base-build (a shell from a landlord partner in Dubai Design District or Expo City) or one adaptive-reuse site (Al Shindagha, Al Fahidi, Sharjah's Bank Street); plan, section and sequence produced with Track 04 tools; feasibility and handover schedules as graded documents
Resources	Pallasmaa (The Eyes of the Skin); Zumthor (Atmospheres); Bachelard; Frampton (Studies in Tectonic Culture); Tschumi; Koolhaas/AMO (Al Manakh, on the Gulf); Ching (Building Construction Illustrated); Neufert; Gehl (Cities for People); Lefebvre; Elsheshtawy, Reisz, Kanna on Gulf urbanism; Sharjah Architecture Triennial publications; Gulf climate data; base-build drawings under NDA; guest critics from regional practices (waiwai, Dabbagh Architects, dxb.lab, X-Architects, Archidentity are examples of the calibre needed)
Evidence	Programmed and choreographed spatial proposal: adjacency map, plans and sections, sequence storyboard, atmospheric strategy (light, thermal, acoustic, olfactory), BOH and egress diagram, fit-out economics sheet, handover schedule set

Unit	Teach as	Interactive
3.2.1 Lineage and regional modernisms	Phenomenology readings; courtyard, sikka, barjeel and mashrabiya as climatic technology; post-oil urbanism	Courtyard-to-Tower Morph (typology timeline); audio-guided Phenomenology Walk with a recording template
3.2.2 Programmatic architecture and scenography	Bubble diagrams; adjacency matrices; hybrid-typology sprints; DNA-to-space translation	Program Adjacency Builder (friction, privacy, acoustic scores); Typology Hybridiser; Brand-to-Space Translator
3.2.3 Choreography	Sequence storyboards; section studies; compression and release	Spatial Sequence Choreographer (draw a path, set volumes, read the tension curve); Threshold Designer
3.2.4 Adaptive reuse	Palimpsest reading of a real building; keep, strip, overwrite	Palimpsest Layer Tool; Retrofit Risk Checklist
3.2.5 Tectonics and fabrication	1:5 joint studies; reveal and shadow-gap detailing; intro to robotic fabrication (with 4.10)	Detail Anatomy Explorer (exploded joints); Patina and Maintenance Timeline
3.2.6 Atmosphere and microclimate	Sun path, shading, thermal mass, evaporative and wind-driven cooling for 25°N; lighting logic; scent and touch strategy sheets	Gulf Climate Sandbox; Motivated Lighting Studio (day to night); Acoustic Character Simulator (reverberation by geometry and material)
3.2.7 Urban interface	Edge-condition studies; setbacks, plinths, canopies; zoning literacy with municipal examples	Edge Condition Configurator (activation score); Context Massing Tool

Unit	Teach as	Interactive
3.2.8 Invisible architecture	BOH mapping; egress diagrams; MEP coordination as literacy	Back-of-House Flow Puzzle (route staff, deliveries and waste without crossing guests); simplified Egress Checker; Ceiling Coordination Puzzle
3.2.9 Spatial economics	CapEx vs OpEx over ten years; yield per square metre; lease-line rules	Fit-Out Economics Calculator; Lease-Line Constraint Overlay
3.2.10 Representation and digital twins	Plan, section, elevation discipline (with 4.1); schedules; BIM and walkthrough (with 4.5)	Drawing Error Hunt; Schedule Builder (lighting, hardware, finishes); Digital Twin Walkthrough template

3.3 Object, Product & Furniture

Dimension	Content
Essential question	How does a thing earn the right to exist: for the hand, the room, the factory, the shipping container and the recycling stream?
What is really being taught	The object as agent, ergonomics at the micro scale, prototyping as inquiry, the economics of the BOM and the box, and end-of-life as a design brief; the most kit-teachable studio
Online strategy	A home kit (cardboard, blue foam, foam densities, fasteners, callipers, scale, a microcontroller for 3.3.8); a resolved object family taken from buck to BOM; a regional craft translation exercise with a documented consultation

Dimension	Content
Resources	Norman; Sennett (The Craftsman); Adamson (Thinking Through Craft); Chapman; Papanek; Munari; Mari (Autoprogettazione); Fukasawa and Morrison (Super Normal); Lefteri (Making It); Thompson (Manufacturing Processes for Design Professionals); Ashby (Materials and Design); Ellen MacArthur circular-design guides; regional craft partners (Irthi Contemporary Crafts Council; Tashkeel's design programmes); regional designers as case studies and critics (Aljoud Lootah, Khalid Shafar, Abdalla Almulla, Ammar Kalo, Nada Debs, Bokja, Ayah Al Bitar)
Evidence	Object family: bucks, prototypes, ergonomic prove-out, process route, BOM and margin sheet, compliance path, flat-pack study, disassembly and end-of-life map

Unit	Teach as	Interactive
3.3.1 Ontology, critical design, craft	Object biographies; critical-design case studies; craft translation with consultation	Object Biography Builder; Craft-to-Industry Ladder (systemic vs extractive check)
3.3.2 Affordance and ergonomics	Measure your own body and kitchen; grip and reach tests; haptic detailing cases	Anthropometric Fit Tester (percentiles against a design); Micro-Interaction Detailer (click, resistance, weight); Thermal Touch Chart
3.3.3 Typology and proxemics	Archetype deconstruction (continues 1.2); use-choreography scripts; room-gravity studies	Use Choreography Scriptor; Room Gravity Planner (circulation and sightlines); Kinetic State Explorer
3.3.4 Prototyping and the buck	Cardboard and foam bucks; digital-analog loops; prove-out on real people	Buck Planner (question, fidelity); Prove-Out Recorder (pressure points on a body map)

Unit	Teach as	Interactive
3.3.5 Soft architecture	Foam layering; webbing; pattern for 3D volume	Comfort Stack Simulator (densities to sink and recovery curves); Upholstery Pattern Unfolder
3.3.6 Industrial methods and honesty	Process selection; tolerances; material-truth audits	Process Selector (volume by material by geometry to routes and cost curves); Tolerance Stack Visualiser
3.3.7 Compliance and safety	Load-test literacy; fire and toxicity codes; certification awareness	Compliance Path Finder (awareness level); Virtual Load Test (simplified stress visual)
3.3.8 Cyber-physical hybridity	Cavities and thermal management; physical UI states; parametric form (with 4.3, 4.7)	Embedded Component Fit; State-Indicator Designer
3.3.9 Supply chain and economics	Flat-pack logic; BOM; value engineering; packaging as threshold	Flat-Pack Volumetric Optimiser; BOM and Margin Calculator that flags DNA violations; Unboxing Storyboard
3.3.10 Circularity	Disassembly maps; patina materials; end-of-life routing	Disassembly Map; End-of-Life Router; Planetary Weight level 2

3.4 Body, Fashion & Wearables

Dimension	Content
Essential question	What does a garment do to a body in motion, to a social reading, to a factory, and to the people and land that made it?

Dimension	Content
What is really being taught	Soft architecture in motion, textile physics, pattern tectonics, the semiotics of dress with genuine cultural authority (modesty as architecture, regional dress as climatic technology), collection mathematics, and the tech-pack as legal document; the strongest market rationale of the four studios
Online strategy	A half-scale dress form, calico, pattern paper and a swatch kit at home; CLO3D or similar on educational licence for drape and simulation; a capsule of hero, core and carry-over pieces; grading and tech-packs are spreadsheet-native and ideal for interactives
Resources	Evans (Fashion at the Edge); Bolton (Rei Kawakubo/Comme des Garçons: Art of the In-Between); Wilson (Adorned in Dreams); Hebdige (Subculture); Lewis (Muslim Fashion); Bucar (Pious Fashion); Fletcher (Sustainable Fashion and Textiles); Thomas (Fashionopolis); Aldrich (Metric Pattern Cutting); Kiisel (Draping); regional textile and dress histories (needs a commissioned reader); regional institutions and designers as partners and critics (Fashion Trust Arabia, the Saudi Fashion Commission; designers such as Faiza Bouguessa, Rami Al Ali, Mohammed Ashi, Arwa Al Banawi, Roni Helou)
Evidence	Capsule collection: silhouette studies, pattern and drape simulation, construction map, grading logic, range plan, tech-pack, campaign frame, labour and material ledger

Unit	Teach as	Interactive
3.4.1 Lineage, historiography, theory	Silhouette-and-power timeline; deconstruction case studies; trickle-up; regional dress as technology; modesty as architecture	Silhouette Power Timeline; Deconstruction Move Library; Regional Garment Anatomy (the kandura and abaya as airflow and layer systems)
3.4.2 Somatic architecture	Ease calculations; motion tests on yourself; volume manipulation	Ease Calculator; Silhouette Distorter on an avatar; Motion Drape Preview
3.4.3 Textile epistemology	Swatch kit (grain, bias, knit); bio-material cases; embellishment as structure	Grain and Bias Simulator; Textile Behaviour Deck; Embellishment-as-Structure Explorer
3.4.4 Pattern tectonics	Flat drafting basics; draping on the half-scale form; seam architecture; closures	Pattern Unfolder (3D to 2D and back); Seam Architecture Selector; Closure Engineering Picker
3.4.5 Wearable technology	E-textile basics (with 4.7); microclimate garments for 45 °C summers as the brief	Heat-Stress Garment Simulator (ventilation, phase-change sliders); E-Textile Circuit Sketcher
3.4.6 Semiotics of dress	Armour and vulnerability readings; code subversion; regional vernacular with authority	Dress-Code Reader (change an attribute, watch the social reading shift); Modesty Volume Studio
3.4.7 Inclusive and adaptive	Adaptive closures; seated drafting; non-linear grading	Grading Logic Visualiser (linear vs proportional grading breaking a silhouette); Adaptive Feature Picker
3.4.8 Collection architecture	Hero, core, carry-over matrix; range plan; drop sequencing (with 2.3)	Range Plan Matrix with margin and press-value weights; Outfitting Logic Checker
3.4.9 Digital twins and XR	CLO3D intro; digital-only garments and their economics	Cloth Physics Playground (simplified); Digital Skin Economics Calculator

Unit	Teach as	Interactive
3.4.10 Editorial and performative release	Styling as design; art-direction briefs; the runway as installation (with 3.1)	Styling Variation Lab; Runway Choreographer (set, sound, order, timing)
3.4.11 Industrial translation and tech-pack	Technical flats; BOM; measurement charts; QC tolerances	Tech-Pack Builder with Error Hunt (find the spec a factory would misread); Shrinkage and Tolerance Calculator
3.4.12 Supply chain ethics and circularity	Labour-chain tracing; regenerative sourcing; take-back design	Garment Cost and Labour Ledger (who earned what); Mono-Material Recyclability Checker; Planetary Weight level 2

Track 04 — Technical & Computational Studios

Teach tools just in time, attached to the module they serve, as drills plus one task on the learner's own project; teach the computational thinking as durable content and keep software-specific steps in swappable tool cards, because every tool named here will change before the program is two years old.

Core toolkit (everyone): 4.1 drawing and notation, 4.3 parametrics, 4.4 machine collaboration, 4.5 real-time and interface, 4.8 rendering and sound. **Electives (choose two):** 4.2 reality capture, 4.6 data visualisation, 4.7 physical computing, 4.9 environmental analytics, 4.10 fabrication, 4.11 data ethics. Version control moves from 4.11 into the core, because the DNA is versioned from 2.2 onward.

Module	Essential question	Teach as	Interactive	Evidence
4.1 Analog Intelligence & Notational Logic (core)	Can you see a thing clearly enough to draw how it is made?	Daily 15-minute drills (gesture, line weight, silhouette); orthographic sets of a real object; storyboard and kinetic notation of a mechanism; paper bucks	Orthographic Trainer (rotate a 3D object, draw its plan/section/elevation, get overlaid feedback); Line-Weight Coach; Movement Notation Sketcher	Orthographic set and movement notation of one existing object or space

Module	Essential question	Teach as	Interactive	Evidence
4.3 Algorithmic Geometry, Scripting & Parametrics (core)	What single rule generates the whole family?	Topology basics (NURBS vs mesh, watertightness); node-based logic; one creative-coding sketch; form-finding by simulated forces	Node Playground (browser-based visual scripting on a simple geometry); Family Generator (one slider regenerates a range); Force Form-Finder (gravity, tension, load)	A parametric definition where one variable regenerates a coherent family
4.4 Synthetic Workflows & Machine Collaboration (core)	Can a machine build from your DNA without drifting into the generic?	Prompt architecture; agent-team briefing; multimodal generation constrained by the DNA; adherence scoring; folder and workflow architecture (ties to 5.4)	Prompt Architecture Studio with DNA Adherence Scorer; Agent Team Builder (roles, briefs, handoffs); Drift Detector (generic vs on-DNA output comparison)	Prompt architecture and agent brief that reproduce the DNA under blind interpretation
4.5 Spatial Computing, Simulation & Interface (core)	Can someone walk through the world before it exists?	Real-time engine basics; a walkthrough of the studio project; collision and trigger logic; a clickable interface prototype	Walkthrough Template (import a model, set lighting and triggers); Interaction Trigger Sandbox; Interface Prototype Kit	A real-time walkthrough or interactive prototype of the studio project
4.8 Optical Physics, Sonic Engineering & Visual Evidence (core)	Does the picture tell the truth about scale, material and cost?	PBR materials; camera, focal length and motivated light; a DAW and spatial-audio basics; a post-production ethics checklist	PBR Material Lab (roughness, metalness, subsurface); Cinematography Coach (focal length and scale cues); Spatial Audio Mixer; Render Honesty Audit	Rendered and scored evidence set with a representation-ethics statement
4.2 Reality Capture & Reverse Engineering (elective)	What is actually there, to the centimetre?	Phone photogrammetry of a real object; a point cloud of a small site; cleaning and decimation	Point Cloud Explorer; Photogrammetry Quality Checker	A capture of a real site or artifact, cleaned and usable

Module	Essential question	Teach as	Interactive	Evidence
4.6 Information Architecture & Data Visualization (elective)	How do you make research legible enough to act on?	Parse and clean a dataset; choose the right encoding; build one interactive view; spatialise one dataset	Encoding Chooser (data shape to chart type); Node-Map Builder; Data-to-Space Mapper	An interactive visualisation of the learner's research data
4.7 Physical Computing & Mechatronics (elective)	How does an object sense and respond?	Microcontroller kit; one sensor, one actuator, one behaviour; biometric input as brief	Circuit Sketcher; Sensor-to-Actuator Logic Builder; Behaviour State Machine	A sensing and actuating prototype
4.9 Environmental Analytics & Lifecycle Computation (elective)	What does the design cost the planet, in numbers?	Sun, wind and thermal simulation of the studio project; an LCA with a real database	Climate Simulation Lite (solar, shading, ventilation for the Gulf); LCA Calculator level 3 (embodied carbon, water, toxicity)	LCA and climatic simulation of the studio project
4.10 Digital Fabrication & Material Mechanics (elective)	Will the machine make what the model shows?	Toolpath and G-code literacy; additive constraints (bridging, supports, shrinkage); shear and fatigue basics	Toolpath Previewer; Print Failure Predictor; Failure Analysis Cases	A fabricated component with its toolpath and a failure analysis
4.11 Data Architecture, Cyber-Ethics & Version Control (elective; versioning to core)	Who owns the data your object collects, and can you roll back a bad decision?	Git-style versioning of the DNA; BIM and PLM as data-rich models; bias and surveillance review of a sensor-heavy design	DNA Repository with branches and diffs; Ethics Review Checklist for IoT and wearables	A versioned DNA repository with an ethics review

Software to budget for (candidates to verify at build time; prefer free or educational licences): Blender and Rhino with Grasshopper for 4.3; CLO3D or Marvelous Designer for 3.4; Unreal or Unity, and TouchDesigner for 4.5 and 3.1; p5.js and Python for creative coding; Arduino for 4.7; a DAW such as Reaper with spatial-audio plugins for 4.8; a photogrammetry app for 4.2; Ladybug Tools for climate and openLCA or a commercial LCA

tool for 4.9; Git for versioning; current image, video and 3D generation models for 4.4, accessed through the program rather than individual accounts so outputs can be logged and scored.

Tracks 05 + 06 — Venture, Release & Defence

Merge the two tracks into one, taught after the primary studio so that every money, law and logistics question is asked about the learner's real artifact; teach finance and law as decision literacy (simulations, not advice), and make the capstone one event: a transfer brief, a live defence and a dossier.

Why merge. 5.8 and 6.1 are the same text; 5.10 is 6.4 plus 6.5; 6.2 restates 3.2.10, 3.4.11 and 5.3. A learner who meets the same paragraph twice concludes the program was assembled, not designed. The merged track has twelve modules and about 150 hours.

Regional specifics to build in (and to have reviewed by counsel and an accountant before release): free-zone versus mainland licensing in the UAE and its consequences for ownership and trade; IP registration routes for the UAE, GCC and international filings; the actual funding landscape for creative practice in the Gulf (institutional production grants, foundation patronage, brand commissions, national creative-economy programs) with current names verified at build time; import regulations and cross-border logistics for Saudi Arabia and the wider GCC.

Module	Essential question	Teach as	Interactive	Evidence
5.1 Masterbrand Strategy & Orbital Architecture	What is the vehicle that carries the world, and how far can a sub-brand drift from it?	Brand-architecture case dissections (monolithic, endorsed, house of brands); drafting non-negotiable codes; ecosystem mapping across verticals	Orbital Architecture Mapper (distance, hierarchy, rules of engagement per satellite); Brand Code Stress Test (apply codes to five touchpoints)	Brand architecture map for the learner's world

Module	Essential question	Teach as	Interactive	Evidence
5.2 Capital Structuring & Creative Finance	Whose money, on what terms, and how long until you run out?	Term-sheet anatomy; dilution mechanics; grant and PPP landscape; cash-flow modelling with receivables versus supplier deposits	Runway and Burn Simulator (18 months, two scenarios); Dilution Calculator (angel, seed, series); Grant Fit Finder (criteria versus DNA compromises)	18-month cash-flow model with a bootstrapped and a funded scenario
5.3 Legal Architecture, IP & Cultural Sovereignty	What can you own, what must you credit, and what will you do when copied?	IP taxonomy (trademark, design right, patent, copyright); contract anatomy (NDA, rights transfer, work-for-hire); attribution and consultation protocol (from 1.1); plagiarism response plans	IP Decision Tree (what protects what, where, at what cost); Contract Clause Explorer (rights transfer, moral rights, credit); Copy Response Playbook	IP and attribution protocol for the project
5.4 Agency Operating System & Studio Governance	Can the studio run without you in the room?	Folder and file architecture; agent teams and prompt templates (with 4.4); pitch as defence; specialist vetting	Studio OS Kit (folder tree, naming, versioning, handover checklist); Pitch Architecture Builder; Specialist Vetting Scorecard	Studio operating system and a pitch structure

Module	Essential question	Teach as	Interactive	Evidence
5.5 Omnichannel Commerce & Spatial Retail	Where does the transaction actually happen, and what is the store for?	Checkout dissections of luxury and high-conversion platforms; O2O journey mapping; flagship-as-media briefs	Checkout Flow Dissector (steps, friction, drop-off); O2O Journey Mapper; Media-Value Calculator for a flagship	O2O journey map and flagship-as-media brief
5.6 Community Governance & Cult Brand Dynamics	Who gets in, how do they recognise each other, and how much do you let them steer?	Access-tier design; ritual and signifier design; co-creation channel structures	Access Tier Designer (exclusivity versus reach curve); Ritual and Signifier Board; Feedback Loop Architect	Access-tier and ritual design
5.7 Narrative-Led Campaigns & Activation	Does the campaign extend the world or paint over it?	Campaign logic from the World Premise; bilingual copy evaluation; activation mechanics (registration, tracking, journey)	Campaign-to-Premise Checker; Bilingual Copy Evaluator (Arabic and English tone, semiotic fit, register); Activation Workflow Builder	Bilingual campaign architecture and activation workflow

Module	Essential question	Teach as	Interactive	Evidence
5.8/6.1 Implementation Logistics & Collaboration	What is the smallest version that is still the thing?	Collaboration matrix; dependency and risk mapping; MVI negotiation; phased drops against supply realities	Collaboration Matrix Builder; MVI Trade-Off Negotiator (cut cost or time, see which DNA constants break); Drop Sequencer (shared with 2.3)	Collaboration matrix, MVI record, phased drop plan
5.9 Venture Economics & Expansion	What does it cost, what is it worth, and where next?	COGS to price architecture; margin stacks by channel; one-market expansion research plan	Margin Architecture Calculator; Expansion Plan Canvas (regulation, behaviour, logistics per market)	Margin architecture and a one-market expansion plan
6.2 Technical Handovers & Compliance	Could a stranger build it exactly, legally and safely?	The shared Handover Grammar unit: specification, schedules, BOM, tech-pack and spatial pack; compliance checklists by studio	Handover Package Auditor (peer checks a pack for the errors a maker would make); Compliance Path Finder (shared with 3.3.7)	Complete handover package audited by a peer

Module	Essential question	Teach as	Interactive	Evidence
6.3 Responsible Release & Lifecycle Operations	Who maintains it, who repairs it, and how does it die?	Release, maintenance and repair budgets; packaging and distribution as threshold; end-of-life routing	Lifecycle Operations Planner; Planetary Weight level 4 (the final ledger)	Release, maintenance and end-of-life plan
5.10/6.4/6.5 Systemic Transfer, Critical Defence & Dossier (capstone)	Does the method survive a brief you did not choose, in front of people who disagree with you?	A transfer brief (the DNA applied to an unfamiliar constraint, context or collaborator); a live defence with a structured panel; dossier curation with abandoned directions and a future-practice plan	Transfer Brief Generator (random constraint draw); Defence Simulator (AI panel of three critics with different agendas, then a live panel); Dossier Builder; Specialist Gap Finder (shared with 2.4)	Transfer brief, defence recording and rubric, professional dossier

Interactive pattern library

The roughly 100 interactives named in §3–§7 collapse into 25 reusable patterns; build the patterns once as components, then instantiate them per module with content, and make every one of them export an artifact into the learner's Concept DNA or project file, so nothing is a quiz.

Design rules for every interactive

- Do, see, explain: the learner acts first, sees a consequence, then reads the concept.
- Peer overlay: wherever a judgement is made (a sort, a threshold, a rating), show the cohort's distribution afterwards; disagreement is the teaching moment.

- AI critic hook: any artifact produced can be sent to a critic persona briefed on the learner's own DNA.
- Export: each interactive writes to the project (DNA field, evidence file, ledger entry, revision log).
- Bilingual from day one: RTL layout and Arabic type rendering are platform requirements, not a later phase.
- Mobile for reading and sorting; desktop for 3D, drawing and building.

#	Pattern	What it does	Instances (modules)	Stack	Size	Priority
1	Structured Editor with schema, versioning and export (the DNA Builder)	Guided form on the DNA schema; thresholds as sliders; grammar coverage radar; diff between versions; export as brief and as agent prompt	2.2 DNA Builder; 2.1 World Perimeter Card; 5.1 Orbital Mapper; 5.4 Studio OS Kit	React, JSON schema, diff engine, DB	L	1
2	Generative Lab (AI outputs rated for adherence)	DNA or prompt in, N outputs out, learner rates, misreads flagged, adherence charted across versions	2.2 Blind Interpretation; 4.4 Prompt Studio, Drift Detector; 6.4 Defence Simulator panel	LLM and image-generation APIs behind a proxy, rating UI, charts	L	1
3	Spectrum or Card Sort with peer overlay	Place cases on a scale or into bins; reveal expert ruling and cohort heatmap	1.1 Reference-to-Appropriation; 2.2 Boundary Test Arena; 3.4.6 Dress-Code Reader; 3.1.1 Container Neutraliser	React, shared DB	S	1
4	Guess-and-Reveal (predict, then evidence)	Annotate an image or document with a prediction or an error, then see the answer	1.3 Affordance Test; 3.2.10 Drawing Error Hunt; 3.4.11 Tech-Pack Error Hunt; 6.2 Handover Auditor; 4.8 Render Honesty Audit	React, image annotation	S	1

#	Pattern	What it does	Instances (modules)	Stack	Size	Priority
5	Calculator and model	Inputs to outputs with a visible formula and a ten-year or 18-month horizon	3.2.9 Fit-Out Economics; 3.3.9 BOM and Margin; 5.2 Runway, Dilution; 5.9 Margin; Planetary Weight levels 1-4; 3.4.2 Ease; 3.4.11 Shrinkage; 3.1.8 Ephemeral Debt	React, formula engine	S-M	1
6	Systems and adjacency map	Nodes and typed links with scores (feedback, friction, privacy, dependency)	1.5 Ecosystem Map; 3.2.2 Program Adjacency; 5.8 Collaboration Matrix; 5.6 Feedback Loop Architect	React Flow	M	1
7	Matrix and grid planner with heat overlays	Roles by families or tiers by reach with cost, risk, margin overlays	2.3 Collection Matrix; 3.4.8 Range Plan; 5.6 Access Tier; 5.5 Checkout Dissector	React grid	M	1
8	Portfolio, revision history and dossier	Timeline of DNA versions against artifacts; abandoned-directions archive; hypothesis log; final dossier	2.4 Revision Viewer, Abandoned Archive; 2.1 Hypothesis Log; 6.5 Dossier Builder	React, DB, file storage	M	1
9	Parametric Morph with identity meter	Sliders alter a form's structural DNA; peers mark where it stops being itself	1.2 Typology Dismantler; 3.3.3 Kinetic State Explorer; 3.4.2 Silhouette Distorter; 4.3 Family Generator	three.js or p5.js	M-L	2

#	Pattern	What it does	Instances (modules)	Stack	Size	Priority
10	3D Sandbox (light, sound, scale, material)	A scene with placeable lights, sources, figures and PBR materials	1.3 Light Logic; 3.1.3 Scale Power; 3.1.4 Light-as-Mass, Ganzfeld; 3.1.5 Sonic Mixer; 3.2.6 Lighting Studio, Acoustic Simulator; 4.5 Walkthrough; 4.8 PBR Lab	three.js or React Three Fiber, Web Audio API	L	2
11	Sequence and choreography timeline	A path with volumes, events and a tension curve	1.3 Threshold Sequencer; 3.2.3 Spatial Sequence Choreographer; 3.1.7 Audience Ritual Builder; 3.4.10 Runway Choreographer	React, SVG	M	2
12	Role-play negotiation with an AI counterpart	A persona with an agenda, a state, and a score for what the learner conceded	3.1.8 Sponsor Negotiation; 5.8 MVI Negotiator; 5.2 Grant Fit; 6.4 Defence Simulator	LLM with persona and scoring	M	2
13	Synthesis and clustering board	Sticky-note canvas with tags, credibility grades and an insight-statement generator	2.1 Research Synthesis Board; 5.7 Ritual and Signifier Board	React canvas (tldraw-class)	M	2
14	Card deck and library	Browsable, filterable reference cards with sliders or comparisons	1.4 Material Cards; 3.4.3 Textile Deck; 3.4.1 Deconstruction Moves; 3.2.5 Detail Anatomy	React, CMS	S	2

#	Pattern	What it does	Instances (modules)	Stack	Size	Priority
15	Decision tree and path finder	Answer questions, receive a route with costs and caveats	5.3 IP Decision Tree; 3.3.7 and 6.2 Compliance Path Finder; 3.3.6 Process Selector; 4.6 Encoding Chooser	JSON tree, React	S	2
16	Bilingual text tool	Arabic-Latin pairing, hierarchy, mirroring, tone and register evaluation	1.4 Bilingual Text Architecture; 5.7 Bilingual Copy Evaluator	React, Arabic type rendering, LLM evaluator	M	2
17	Layered timeline with counter-layers	Drag lineages; toggle dominant and obscured layers	1.1 Counter-Narrative; 3.4.1 Silhouette Power; 3.2.1 Courtyard-to-Tower	React, d3	M	3
18	Layer and time tool (palimpsest, patina)	Image layers with keep/strip/overwrite; surfaces aged at 1, 5, 20 years	3.2.4 Palimpsest; 1.4 Patina Time Machine, Sign-Swap	React canvas	M	3
19	Scenario generator	Variables in, structured scenarios or briefs out	1.5 Futures Wheel; 3.1.2 What-If; 6.4 Transfer Brief Generator	Templated LLM, forms	S	3
20	Journey and flow mapper	Stages, touchpoints, drop-offs	5.5 O2O Journey; 5.7 Activation Workflow; 3.3.3 Use Choreography; 3.3.9 Unboxing Storyboard	React Flow	S-M	3
21	Routing and fit puzzle	Route flows or fit components under rules	3.2.8 BOH Flow, Egress, Ceiling Coordination; 3.3.8 Component Fit; 3.3.9 Flat-Pack Optimiser	React canvas grid, rules	M	3

#	Pattern	What it does	Instances (modules)	Stack	Size	Priority
22	Node and visual-scripting playground	Small node graphs that evaluate live on geometry or hardware logic	4.3 Node Playground; 4.7 Sensor-to-Actuator; 4.6 Node-Map	React Flow, evaluator	L	3
23	Drawing trainer with overlay feedback	Draw over or beside a reference, get an overlay comparison	4.1 Orthographic Trainer, Line-Weight Coach	Canvas, image comparison	L	3
24	Simplified physics (cloth, foam, forces)	Verlet or spring simulations of drape, sink and load	3.4.3 Grain and Bias; 3.4.9 Cloth Playground; 3.3.5 Comfort Stack; 4.3 Force Form-Finder	three.js, physics library	L	4 (defer to CLO3D and Grasshopper where licensed)
25	Unfolder (3D to 2D)	Pattern and nesting from simple 3D forms	3.4.4 Pattern Unfolder; 3.3.5 Upholstery Unfolder	three.js, custom geometry	L	4

Platform stack (proposal). React and TypeScript; three.js for 3D; Web Audio for sound; React Flow and d3 for maps and timelines; a hosted Postgres with auth and storage (Supabase-class) for projects, cohorts and peer overlays; LLM and image APIs behind a logging proxy so every generative run is attributable and scoreable; an LMS shell or LTI 1.3 integration if the program sells through an institution; RTL and Arabic typography in the design system from the first component.

Sizing. Priority 1 patterns (eight) are enough to ship Track 02 as a pilot. Rough build effort per pattern: S about 1–2 weeks, M about 3–5 weeks, L about 6–10 weeks for a small team, before content instantiation.

Teaching methodology

The program should be built with backward design and constructive alignment (outcome → evidence → activity), delivered as a studio translated online (brief, iteration, pin-up, crit), and assessed on a versioned portfolio rather than tests; the frameworks below are the ones to write into the design brief for every module.

Frameworks to adopt

Framework	What it gives this program	Where it shows up
Backward design (Wiggins and McTighe) and constructive alignment (Biggs)	Every unit starts from an essential question, names its evidence, and only then chooses content and interactives	The Essential question and Evidence rows in §3–§7
Studio as signature pedagogy (Shulman; Schön's reflective practitioner)	Learning happens by making and by public critique of the making; the tutor is a critic, not a lecturer	Fortnightly live crits; async pin-up gallery; the AI critic between
Threshold concepts (Meyer and Land)	A short list of ideas that, once grasped, change how the learner sees everything; teach these slowly, with troublesome cases	The list below
Cognitive apprenticeship (Collins, Brown and Newman)	Model, coach, scaffold, fade: worked examples first, then the learner's project, then independence	Worked exemplar per module (one full case walked through), then the learner's own instance
Experiential cycle (Kolb)	Do, observe, conceptualise, try again; the unit anatomy follows this order	Lab first, reading second
Community of inquiry (Garrison)	Online cohorts need social, cognitive and teaching presence engineered, not assumed	Peer pods of four or five; named critics; structured feedback
Mastery and re-submission	Studio work is pass, revise or resubmit against a rubric, not a percentage	Rubric below; two resubmissions allowed

Threshold concepts of this program (the ideas the whole curriculum turns on)

1. Design history is a web with obscured strands, not a line (1.1).
2. An archetype has a structural DNA you can push far but cannot delete (1.2).
3. The body, not the eye, is the metric (1.3).

4. Materials and words are signs before they are surfaces (1.4).
5. A concept is a generative system with rules, thresholds and anti-rules, not a picture (2.2).
6. Generativity is testable: if a stranger or a machine cannot build from it, it is not finished (2.2, 4.4).
7. The collection is designed before the object (2.3).
8. The invisible systems — back of house, bill of materials, cash flow, code — are design material (3.x, 5.x).
9. Circularity is the brief, not the footnote (the Planetary Weight thread).

Anatomy of a learning unit (60–90 minutes, the building block of every module)

Step	Minutes	What happens
Provocation	3	One question or one image that the unit will answer
Lab	15–25	The interactive, before any explanation
Concept	10–15	Short reading or video that names what the lab showed
Case	10	One regional and one global exemplar dissected against the concept
Apply	20–30	The same move on the learner's own project; the interactive exports the result
Overlay	5	The cohort's choices shown against the learner's
Critic	async	AI critic response on the artifact; one peer prompt
Log	2	One line in the revision log: what changed and why

Cohort rhythm. Weekly unlock on Monday; peer pod on Wednesday (45 minutes, four or five learners, a rotating facilitator); live crit or office hours alternating on Fridays; revision log due Sunday. A 12-week studio has six live crits and one specialist Q&A. Self-paced learners get the same units with the AI critic and asynchronous peer overlay, and can join a crit cohort for a fee.

Critique protocols. Use a fixed structure so feedback is teachable: statements of meaning (what the work does), the maker's questions, neutral questions from critics, then opinions only with permission (Lerman's Critical Response Process). Every crit also runs the four boundary tests from 2.2 as a lens: does the piece belong, sit on the edge, fail the DNA, or seduce while being off-DNA. Train the AI critic personas on the same protocol, each with a declared agenda: the Historian (1.1), the Fabricator (3.x, 4.10), the Buyer (2.3, 5.5), the Community (1.1, 5.6), the Planet (the thread), and the Stranger (blind interpretation).

Assessment rubric (every module uses the same five criteria; module briefs specify the evidence)

Criterion	What it looks for	Fails when
Generativity	The DNA or system lets others build correctly from it; tests and blind interpretation documented	Rules only the author can apply
Craft	Resolution appropriate to the phase; command of the medium and the tools used	Fidelity substitutes for thinking
Feasibility	The invisible systems are addressed: body, code, BOM, cash, site	Feasibility is deferred to "later"
Cultural authority	Sources credited; consultation documented; references distinguished from appropriation	Regional forms used as surface
Evidence and reflection	Alternatives and abandoned directions shown; revision history visible; claims supported	Only the final image is submitted

What to avoid. Long recorded lectures (cap videos at eight minutes); quizzes as assessment; a single final submission with no visible history; separating tools from projects; and the temptation to write learning copy in the manifesto register of the source document — keep that voice for briefs and provocations.

Research and resource needs

The largest resource risk is not readings or software but the regional material that does not yet exist in teachable form; six commissioned assets, a rights strategy and a consultation board should be budgeted before any module is scripted.

Commissioned assets (do not exist yet; the program's real intellectual property)

Asset	Serves	Scope
Gulf and Arab-world design history reader with counter-narratives	1.1, 3.2.1, 3.4.1	8–12 commissioned essays plus an annotated chronology; Arabic and English
Regional material archive	1.4, 3.3, 3.2.5	40–60 materials (palm, areesh, sadu, coral stone, gypsum, teak, contemporary composites) with provenance, behaviour, lifecycle, semiotics, photography
Regional colour and light coding study	1.3, 3.2.6	Field study of colour meaning and lighting norms across Gulf, Egyptian and Levantine contexts
Regional dress and textile reader	3.4.1, 3.4.6	Garment anatomies (kandura, abaya, ghutra, thobe, regional textiles) as climatic and cultural systems; trade-route history
Case-study library	all studios, 2.3, 5.x	60–80 dissected cases, half regional, with cleared images, drawings and interviews
Bilingual glossary	all	The program's terms (Concept DNA, anti-DNA, threshold, grammar, carrier, MVI) defined in English and Arabic, with the regional design vocabulary

Core reading list (the canon the per-module lists in §3–§7 draw from; buy library access or negotiate excerpts)

- Context: Escobar, Designs for the Pluriverse; Tunstall, Decolonizing Design; Said, Orientalism; Reisz, Showpiece City; Elsheshtawy, Dubai: Behind an Urban Spectacle; Kanna, Dubai, the City as Corporation; Koolhaas/AMO, Al Manakh
- Form and senses: Ching, Form, Space and Order; Elam, Geometry of Design; Albers, Interaction of Color; Norman, The Design of Everyday Things; Hall, The Hidden Dimension; Pallasmaa, The Eyes of the Skin; Zumthor, Atmospheres; Bachelard, The Poetics of Space
- Material and meaning: Barthes, Mythologies; Forty, Objects of Desire; Chapman, Emotionally Durable Design; Sennett, The Craftsman; Adamson, Thinking Through Craft; Lupton, Thinking with Type
- Systems and futures: Meadows, Thinking in Systems; Bratton, The Stack; Crawford, Atlas of AI; Dunne and Raby, Speculative Everything; Fletcher, Sustainable Fashion and Textiles
- Method: Alexander, A Pattern Language; Salen and Zimmerman, Rules of Play; Kolko, Exposing the Magic of Design; Wolf, Building Imaginary Worlds
- Art and space: O'Doherty, Inside the White Cube; Kwon, One Place After Another; Bishop, Artificial Hells; Bourriaud, Relational Aesthetics; Schafer, The Soundscape; Careri, Walkscapes; Frampton, Studies in Tectonic Culture; Gehl, Cities for People
- Object and body: Papanek, Design for the Real World; Munari, Design as Art; Lefteri, Making It; Ashby, Materials and Design; Evans, Fashion at the Edge; Wilson, Adorned in Dreams; Lewis, Muslim Fashion; Bucar, Pious Fashion; Aldrich, Metric Pattern Cutting

Institutions and archives to approach (partnership, access, guest critics, sites): Sharjah Art Foundation and the Sharjah Architecture Triennial; Art Jameel (Jameel Arts Centre, Hayy Jameel); Alserkal Avenue; Tashkeel; Irthi Contemporary Crafts Council; Louvre Abu Dhabi; NYU Abu Dhabi Art Gallery; Dubai Design District and Dubai Design Week; Misk Art Institute and the Diriyah Biennale Foundation; Fashion Trust Arabia; the Saudi Fashion Commission; Archnet and the Aga Khan Documentation Center; Arab Image Foundation; Barjeel Art Foundation; Khatt Foundation. Verify current programs and contacts at build time.

People

Role	Count	Purpose
Track leads (subject-matter experts)	6	Own the outcomes and exemplars of each track; sign off module scripts

Role	Count	Purpose
Studio guest critics	3–4 per studio	Live crits, recorded case interviews, specialist Q&A
Cultural consultation board	5–7	Review of regional content, attribution, community consultation protocols; paid, credited
Legal and finance reviewers	2	Review 5.2, 5.3, 5.9, 6.2 for accuracy and for what must be framed as literacy, not advice
Arabic editor and type consultant	1–2	Bilingual copy, glossary, RTL design system
Learning experience designer	1–2	Unit anatomy, storyboards, rubrics
Interactive developers	2–3	The pattern library
Video and photography producer	1	Case interviews, making demonstrations, site captures

Software and licences. See §6. Budget educational or floating licences for Rhino, CLO3D, Unreal or Unity, a DAW, an LCA tool; everything else can be free. Generative model access is metered and should be provisioned by the program, not by learners.

Physical kits (approximate cost per learner, shipped): installation maquette kit (figures, foam, acetate, LED pucks) 40–70 USD; object kit (cardboard, blue foam, foam densities, fasteners, callipers, scale) 60–120 USD; fashion kit (half-scale form, calico, pattern paper, swatch set, tools) 150–250 USD; electronics kit (microcontroller, sensors, servos) 50–80 USD. A regional fulfilment partner is needed; customs into Saudi Arabia and the wider GCC should be checked early.

Sites and base-builds. One landlord partner willing to release a shell's drawings under NDA for 3.2; two or three public sites with permission for 3.1 proposals; a Gulf climate dataset (sun path, temperature, humidity, wind) for 3.2.6 and 4.9.

Data. Anthropometric datasets for 1.3 and 3.3.2; an LCA database licence for 4.9; regional census and archival datasets for 2.1 and 4.6, with access terms.

Rights and ethics. An image-licensing budget and a preference for commissioned photography; consent and attribution templates for oral history and community consultation (1.1, 2.1, 3.3.1); a written policy on how learners' generative outputs are stored,

credited and scored (4.4, 4.11).

Gaps, alternative and adjacent topics

The syllabus promises more breadth in Track 01 than Track 03 delivers, and it under-serves the two practices the region commissions most — brand and cultural-district experience; the table lists what is missing, why, and the cheapest honest way to handle each.

Gap or adjacent topic	Why it matters	Handle as
Brand identity, graphic design and typography	1.1 names it; 5.1 and 5.7 depend on it; every studio's output needs a visual identity	New studio 3.5 Identity, Graphic & Typographic Systems (10 units), or scope it out and state that in the program description
Digital product and interface design	1.3 names interfaces; 4.5 builds them; 5.5 needs them	New studio 3.6 Digital Product & Interface, or a 40-hour adjacent-translation option
Exhibition, museum and cultural-district experience	The region's largest commissioner of experiential design (museums, biennials, heritage districts)	A unit inside 3.2 (Programmatic Architecture) plus a capstone track option
Events, expos and large-scale activations	5.7 touches it; Expo-scale work is a regional specialism	A unit inside 5.7 with a real activation case
Service design, hospitality and F&B	3.2.2 hybridises hospitality; hospitality drives Gulf commercial space	A unit inside 3.2 or an elective micro-course
Moving image, sound and performance as practice	1.1 names them; 3.4.10 and 4.8 need them; no studio owns them	Fold into 3.1 (durational practice) and 4.8, and say so
Landscape, desert ecology and water	The region's defining environmental constraint; absent as a subject	A unit in 1.5 and 3.2.6; a thread through Planetary Weight

Gap or adjacent topic	Why it matters	Handle as
Islamic geometry as generative grammar	The most direct regional bridge from 1.2 to 4.3; currently implicit	A named unit shared by 1.2 and 4.3
Arabic typography and bilingual design	Currently one bullet in 1.4 and one in 5.7; it is a differentiator	A full unit in 1.4 and the Bilingual Text tool (pattern 16)
Game design and interactive narrative	World-building's most developed methodology lives here; 2.1 and 2.2 borrow without saying so	Readings and one case in 2.1; a possible elective
Scent and culinary design	3.2.6 names olfaction; hospitality needs it	Case studies in 3.2.6; an elective micro-course
Jewellery and accessories	Sits between 3.3 and 3.4; a large regional market	An adjacent-translation option
Accessibility and inclusive design as a horizontal	Only 3.4.7 and 3.2.8 address it	A thread with a checkpoint in every studio brief
Labour ethics in the Gulf (construction, garments, craft)	Touched in 1.1 and 3.4.12; sensitive and unavoidable	Frame carefully with the consultation board; teach as evidence and protocol, not polemic
Research ethics and consent	2.1 does primary research with people	A short unit before 2.1 with templates
Documentation and photography practice	3.1.7 and 6.5 depend on it	A unit inside 4.8
Client, stakeholder and negotiation skills	Partly in 5.4 and 5.8	The role-play pattern (12) across the venture track
Data law and sovereignty in the UAE and GCC	4.11 names cyber-ethics generically	Regional specifics reviewed by counsel
Writing and argumentation	The capstone is a defence; nothing teaches the writing	A thread: one written argument per track
Synthetic worlds as a medium (not a tool)	3.1.6 and 4.4 point there	A candidate future studio once 3.5 and 3.6 exist

Alternative program shapes worth testing with learners

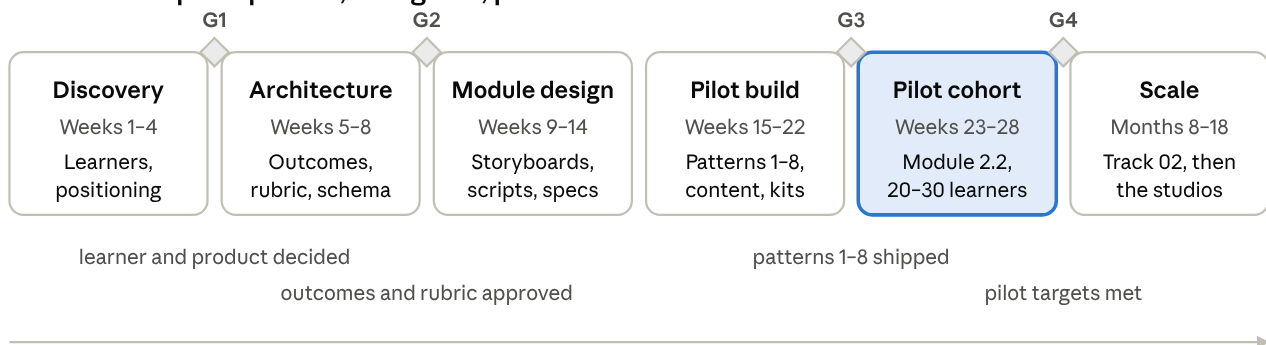
- Studio-first: learners enter a studio in week one and discover the need for a DNA by week four; Track 02 is taught inside the studio. Better for makers, worse for consistency.
- Venture-first: for studio founders, Tracks 05–06 open the program and the studios become optional. Better for revenue, weaker as design education.
- Persona stacks: three micro-credential paths (Practitioner, Founder, Cultural Producer) that share Track 02 and diverge afterwards. Recommended once the pilot has run.

Naming. Replace "Modular" with "Module"; consider renaming the tracks for learners as Foundations, Method, Studios, Toolkit, Venture, and keep "Concept DNA" as the product's proprietary term.

How curricula of this nature are built

Programs like this take about seven months to a paid pilot cohort and 12–18 months to a full catalogue; what separates the ones that ship from the ones that stay syllabi is a single accountable curriculum lead and a gate between every phase that nobody can skip.

Build roadmap: six phases, four gates, pilot in week 23



build roadmap · 6 phases, 4 gates

Each gate is a decision, not a meeting: Discovery cannot end until the learner, product shape and modality are written down; Architecture cannot end until every module has outcomes and the shared rubric is signed by the track leads; the pilot build ends when the eight priority-1 patterns run end to end with real content; the pilot cohort ends with completion and satisfaction numbers that decide whether Scale starts.

What each phase produces

Phase	Deliverables	Owner
Discovery	12–15 learner interviews across the target personas; competitor and price scan; positioning statement; product shape (stackable modules) and modality decision; language decision; commissioned-asset scoping	Curriculum lead with a researcher
Architecture	Program specification (this document, finalised); one module descriptor per module (aims, 3–5 outcomes, evidence, hours, prerequisites); the shared rubric; the Concept DNA schema; the pattern library specification; the Planetary Weight and Handover Grammar threads defined	Curriculum lead, track leads, learning experience designer
Module design (pilot)	For Module 2.2 and its Track 04 companions: unit-by-unit storyboards on the unit anatomy; scripts; interactive specifications with data models and export formats; exemplar case fully worked; AI critic persona briefs	Learning experience designer, instructional designer, track lead for Track 02
Pilot build	Patterns 1–8 as components; 2.2 content instantiated; DNA Builder and Blind Interpretation Lab live; cohort tooling (pods, crits, pin-up gallery); kits not needed for 2.2	Developers, producer, editor, Arabic editor
Pilot cohort	20–30 paid learners over four to six weeks; two live crits; instrumentation on every interactive; exit interviews; a revision list	Curriculum lead, one tutor, one critic

Phase	Deliverables	Owner
Scale	Track 02 complete, then one studio at a time, Track 01 in parallel as self-paced, venture track last; commissioned assets delivered; institutional partnerships and any accreditation route pursued	Whole team

How real programs of this nature work. Studio-based master's programs (RCA, CSM, Design Academy Eindhoven, the Bartlett) are built from a program specification and module descriptors validated by an academic board and read by an external examiner; every module states aims, outcomes, assessment and hours, which is exactly what the source document lacks. Cohort-based online programs (the model behind the better professional design and strategy schools) run a small paid pilot before building the catalogue, treat the live crit as the product, and measure completion rather than enrolment. Interactive learning content is expensive: a planning figure of 40–100 production hours per finished learner hour for interactive units is a common industry rule of thumb, approximate, and the reason to build reusable patterns rather than bespoke pieces.

Team for the pilot (smallest viable): curriculum lead (owner), Track 02 lead, learning experience designer, one full-stack developer plus one 3D or creative-coding developer, editor, Arabic editor, part-time producer, one tutor for the cohort. Add track leads and guest critics per studio in Scale.

Cost drivers to model early: commissioned regional assets (§10); developer time on the L-sized patterns (2, 10, 22–25); generative model usage per learner; kit fulfilment into the GCC; guest critic fees; image and text rights; any accreditation or partnership fees.

How we build it together

The practical division of labour: you make the six decisions in §1 and own the regional voice and the relationships; I draft every document, prototype every interactive pattern as a working page, run the research scans, and revise from your comments in this doc.

What I can produce, concretely

Category	Outputs
Curriculum documents	Finalised program specification; one module descriptor per module (aims, 3–5 outcomes, evidence, hours, prerequisites) for all 39 modules; the shared rubric with level descriptors; unit-by-unit storyboards on the unit anatomy; scripts and on-screen copy in English with Arabic drafts for review; the bilingual glossary; annotated reading lists; case-study dissections from material you supply
The method as software	The Concept DNA schema as a JSON structure with validation rules; export formats (contributor brief, agent prompt); the AI critic persona briefs (Historian, Fabricator, Buyer, Community, Planet, Stranger) and the critique protocol they follow
Interactive prototypes	Working, clickable pages for the priority-1 patterns (DNA Builder, Boundary Test Arena, Blind Interpretation Lab with live model calls, calculators, matrix planner, systems map, guess-and-reveal, revision history), published as private pages you can test with people; three.js prototypes for the 3D sandbox; Figma mockups of the platform shell
Platform groundwork	A database schema for learners, cohorts, projects, DNA versions, artifacts and peer overlays; a build plan for the pattern library; a prompt-logging proxy design for generative runs
Research	Literature scans for each module through the academic search connectors; competitor and price scans; verification of regional institutions, programs and contacts before anything is printed; learner-interview guides and synthesis
Program management	A module tracker (status, owner, outcomes written, storyboard, build, review) as a sheet or a tab in this doc; weekly scheduled check-ins that summarise what moved and what is blocked

Working setup

- This doc stays the program specification; comment on any line and I revise. Decisions get recorded at the top of §1.
- One doc per module descriptor, generated in batches of a track at a time, so your track leads can review in their own area.
- Prototypes live as pages you can share with testers; each one carries its own short test script.
- Assets (readings, images, case material) in a shared folder you control; I read from it rather than you pasting.

Proposed first sprint (four weeks)

- ☐ Week 1 — You answer the six decisions in §1. I deliver three learner personas with an interview guide, a competitor and price scan, the DNA schema as JSON, and the rubric with level descriptors.
- ☐ Week 2 — Module 2.2 descriptor and a ten-unit storyboard; AI critic persona briefs; the module descriptors for Track 02's other three phases in draft.
- ☐ Week 3 — DNA Builder and Boundary Test Arena as working pages; a five-person hallway test; revisions.
- ☐ Week 4 — Blind Interpretation Lab with live model calls; unit scripts for 2.2's first five units; a pilot-cohort plan (recruitment, price, dates, tutor).

What I need from you to start

1. Answers, even provisional, to the six decisions.
2. Any positioning, brand or business-model material that already exists.
3. Two or three examples of work you consider on-DNA for the program's own voice, so the exemplar case in 2.2 is yours rather than borrowed.
4. Names of the people who will be track leads or first critics, if known.
5. A budget envelope and a target date for the pilot cohort, so the roadmap in §12 is fitted rather than generic.

Immediate options. Say the word and I can produce next: the 39 module descriptors as a batch; an Arabic version of this document; the tracker sheet; or the first working page of the DNA Builder.